

FINAL PROJECT – Individual Report [Week 1]

Group Name: JaiJaiSungLui

Group Number: Group 3

Name: Nathi Ritnatikul

Calendar Week: CW40

Date: [Oct 6–12, 2025]

Week 1 Progress:

- Created CAD files for all metal parts to ensure they are listed for future return.
- Designed several versions of the Auto Robot and refined them according to rulebook specifications.
- Finalized the drive frame and cube collection mechanism (gate) design.
- Created an assembly CAD file of the drive frame and the gate mechanism.

Challenges & Solutions:

- **Challenge 1:** The rulebook specifications on auto robot have a limitation on its size and the number of electronics components that can be used.
- **Solution:** We separate the whole robot into each mechanism including drive train, gate mechanism, sensors, and other. Then we assign the electronics to each part so that it is not over the specifications limits.
- **Challenge 2:** To connect the DC motor and Micro servo to the makeblock metal parts, there is a problem on the size of the screw that the M4 screw (Makeblock) does not fit with the electronics components.
- **Solution:** Measured and identified compatible screw sizes (either M3 or M2) for proper fit.

Next Week Plan:

- Purchase the required components to complete the robot's mechanisms.
- Complete the detailed CAD assembly and combine all parts into a single file.
- Begin initial concept design for Manual Robot.